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Novel furo[2,3-*d*]pyrimidinones as direct thrombin inhibitors: Design, synthesis and biological evaluation

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ABSTRACT

Thrombin, a versatile protease is a promising target in coagulation cascade for thrombosis related conditions. We designed and synthesized thirty novel furo[2,3-*d*]pyrimidinone analogs and evaluated them for their antithrombotic potential. Initial *in vitro* screening revealed that most of the compounds displayed good to moderate level of antithrombin properties. Compounds (**7b,** IC_{50} = 0.96 μM and **7g,** IC_{50} = 1.36 μM) displayed the most potent direct thrombin inhibition activity. The results from the *ex vivo* anticoagulant activity demonstrated compound (**7b)** having prolonged PT and aPTT in rats relative to the control. Interestingly, compound (**7b,** BT = 91 s.) exhibited better safety profile than the standard drug dabigatran (BT = 102 s.) in terms of bleeding risk. Molecular docking and dynamics simulation studies identified and validated the binding interactions of potent derivatives with the target protein thrombin.

1. Introduction

New drug development is a challenging process including high costs, time-consuming efforts and high failure rates. With the advances in artificial intelligence (AI)-based and machine learning (ML) methods, researchers could easily explore new target identification, discovery of lead and its optimization, pharmacokinetic and toxicity predictions, synthesis planning, biological testing to clinical trial management.[1] Although AI powered methods are cost-effective and more efficient, they faces significant obstacles like lack of standard and unbiased data, interpretability, undruggable targets and high computational requirements.[2] In this scenario, drug design strategies *e.g.* fragment-based and structure-based drug discovery, scaffold hopping, molecular hybridization, bio-isosterism techniques are explored in combination with AI-based computational methods.[3–8] Several new molecules are developed by mechanism based,[9] gene-based,[10] structure and shape-based[11] design strategies. One of the newly developed chemical biology technologies *viz.* proteolysis targeting chimeras (PROTACs) are designed to selectively target and degrade cancer proteins.[12,13]

Thromboembolic diseases comprising of generation of clot in blood circulation continue to be one of the major reasons for morbidity and mortality today.[14] Treatment with traditional anticoagulants like warfarin or low molecular weight heparins (LMWHs) are responsible for higher bleeding, drug-drug interactions, requirement of dose monitoring, and low therapeutic indices demanding a significant task to antithrombotic therapy.[15,16] Apixaban, rivaroxaban and edoxaban are approved oral antithrombotic factor Xa inhibitors, while dabigatran is the only orally active direct thrombin inhibitor available in the market for the treatment of thrombosis. Some other thrombin inhibitors like argatroban (Fig. 1) and heparin are given parenterally.[17].

Without any doubt, these new direct oral anticoagulant (DOAC) drugs have proved their efficacy but their use is restricted or contraindicated in some circumstances though they offer several advantages over VKAs. Also, the risk of major bleeding remains high.[18] These

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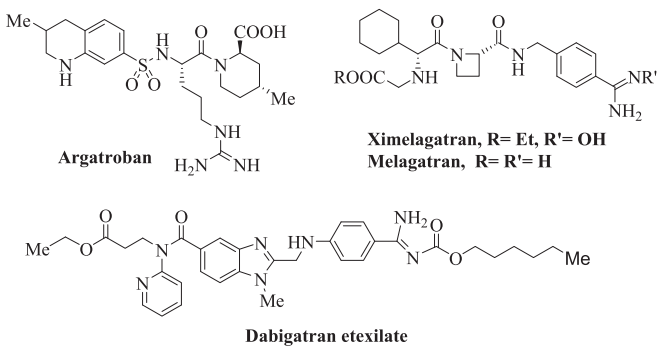


Fig. 1. Early developed direct thrombin inhibitors.[27]

DOACs are not effective in patients having mechanical heart valves, in valvular atrial fibrillation or antiphospholipid syndrome, and should be used with caution or avoided in patients with severe renal or hepatic impairment. Their safety in pregnancy, breastfeeding women and pediatric population remains in question.[19,20] Greater costs for their acquisition and non-availability of effective reversal agents are associated with their poor accessibility and patient non-compliance.[21] Consequently, developing new anticoagulants with safe monitoring and without bleeding tendency is of utmost interest.

Thrombin, a multifunctional serine protease from the coagulation cascade is the widely explored target in the anticoagulation treatment.[22] Due to its crucial role in coagulation and hemostasis, thrombin is projected as a critical target for the development of novel antithrombotic agents.[23] Direct thrombin inhibitors target both soluble and fibrin-bound thrombin and also inhibit the activity of thrombin-mediated factors V, VIII, XII, fibrinogen and platelets.[24] Moreover, heparin-induced thrombocytopenia is not observed for them.[25] Additionally, they can inhibit platelet aggregation and exhibit anti-inflammatory properties, suggesting their wide applications in the treatment of thrombotic disorders.[26]

2. Rational designing of target compounds

Rigorous efforts in the direction of development of antithrombotic agents aiming oral bioavailability, better safety along with high efficacy and selectivity of the developed molecules has been made in the last decades but only a few agents could make it to the market.[27] Dabigatran

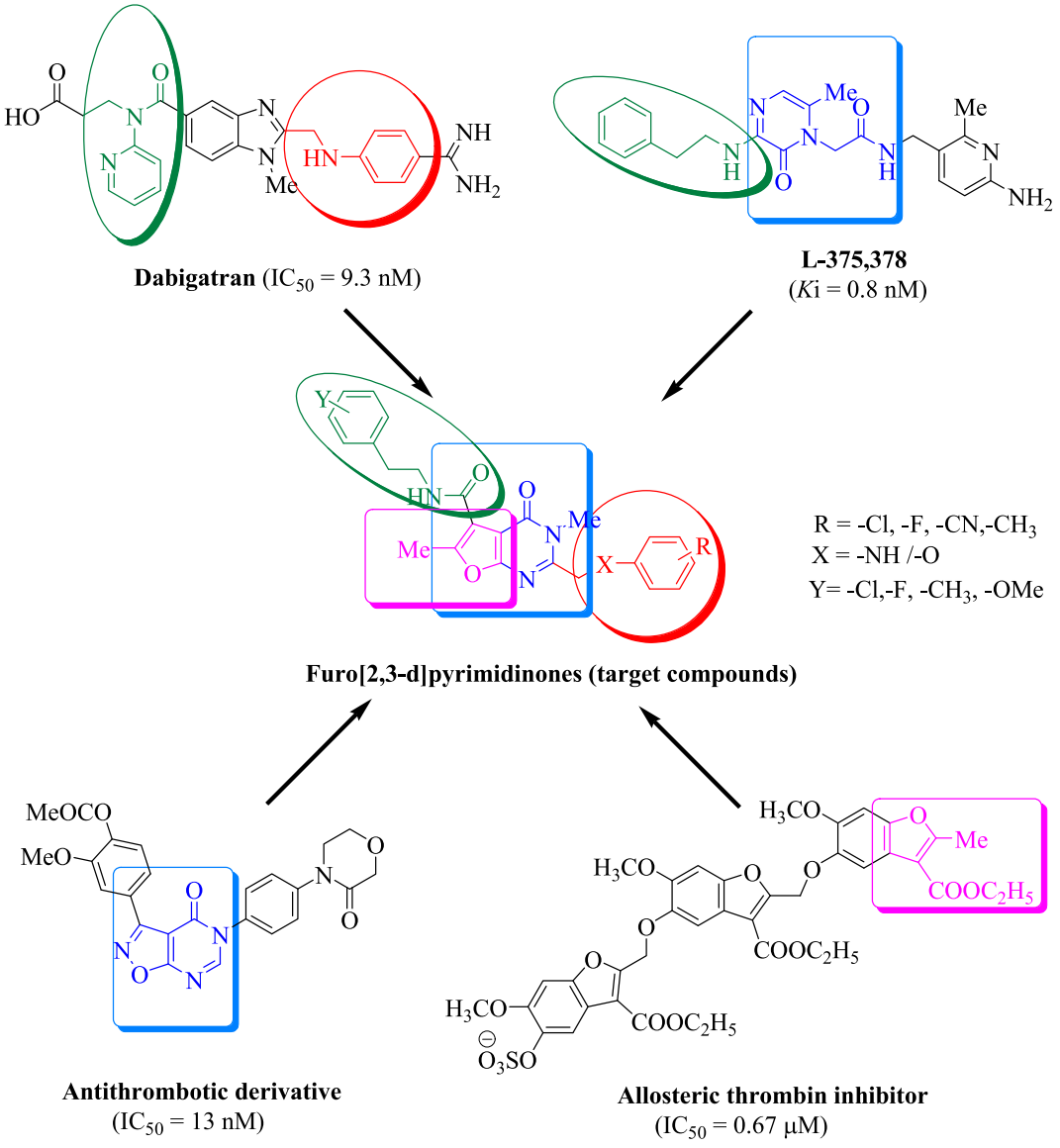


Fig. 2. Designing strategy of the target compounds.[31,36,37,46]

(Fig. 1), the first orally approved direct thrombin inhibitor has very low oral bioavailability and poor pharmacokinetic properties because of the presence of highly basic amidine group in its structure.[28] Dose-dependent hemorrhage and high renal clearance of dabigatran contribute its limited use in thrombosis patients.[29,30] Therefore, search for an ideal direct thrombin inhibitor with acceptable safety and efficacy levels still continues. Several heterocyclic moieties like benzimidazole,[31–33] benzothiophene,[34] benzoxazine,[35] benzofuran,[36] pyrazinone,[37] pyridinone,[38] pyrimidinone,[39] quinoline[40] and quinoxaline[41] have been explored by medicinal chemists to obtain direct thrombin inhibitors. Furanopyrimidine scaffold is reported earlier for various biological activities[42–45] while benzofuran analogs have exhibited allosteric thrombin inhibition.[36] Also pyrazinone[37] and isoxazolopyrimidinone[46] derivatives displayed potent antithrombotic activity and pharmacokinetic properties. Thus, in continuation of our studies on development of novel antithrombotic agents,[47–49] we selected furo[2,3-*d*]pyrimidinone scaffold and biologically active templates employing fragment-based drug design, bioisosterism and molecular hybridization strategies, and designed some novel furo[2,3-*d*]pyrimidone derivatives containing S1-occupying neutral moieties and S3-binding hydrophobic aromatic motifs (Fig. 2, Table 1), with the aim to obtain derivatives possessing improved biological profile.[50]

3. Results and discussion

3.1. Chemistry

The scheme followed for the synthesis of novel furo[2,3-*d*]pyrimidone derivatives is displayed in Scheme 1. The first compound, diethyl 2-amino-5-methylfurandicarboxylate (**3**) was obtained from ethyl cyanoacetate (**1**) and ethyl 2-chlroacetoacetate (**2**) in isopropanol solvent under basic conditions at room temperature.[51] Compound (**3**) on treatment with chloroacetonitrile in presence of dry HCl in dry dioxane solution led to the cyclization product (**4**)[52] that on alkylation from dimethyl sulphate in dry acetone conferred compound (**5**). Nucleophilic substitution of the chloro group in compound (**5**) with substituted anilines or phenols yielded intermediate compounds (**6a-6j**). Finally, the target compounds (**7a-7u; 8a-8i**) were obtained from the treatment of derivatives (**6a-6j**) with different phenethylamines and trimethyl aluminium through Weinreb transamidation.[53,54] Structures of the target compounds were established using ^{1}H NMR, ^{13}C NMR, HR-MS and elemental analyses data before proceeding for biological activity.

3.2. Biological activity

3.2.1. In vitro thrombin enzyme inhibition

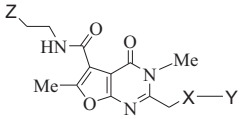
The final compounds were evaluated for *in vitro* thrombin (FIIa) inhibitory activity using standard dabigatran etexilate.[55,56] The outcome of the results revealed that all of the synthesized compounds exhibited more than 50% inhibition at 20 μM concentration (Fig. 3). The compounds that showed >60% inhibition (total 22 compounds) were selected for their calculation of IC_{50} values. The opted compounds from the series exhibited comparable thrombin inhibitory activities ranging IC_{50} values from 0.96 to 68.6 μM (**Table 1**). Compound **7b** containing 4-chlorophenyl and compound **7 g** with 4-fluorophenyl groups as S1-binding moieties were observed as the most potent derivatives with IC_{50} values of 0.96 and 1.36 μM respectively.

To analyze the selectivity of the compounds, a Factor Xa chromogenic assay was performed in which less than 10% inhibition against FXa enzyme was recorded at 20 μM concentration for all these compounds.

3.2.2. Structure-activity relationship studies

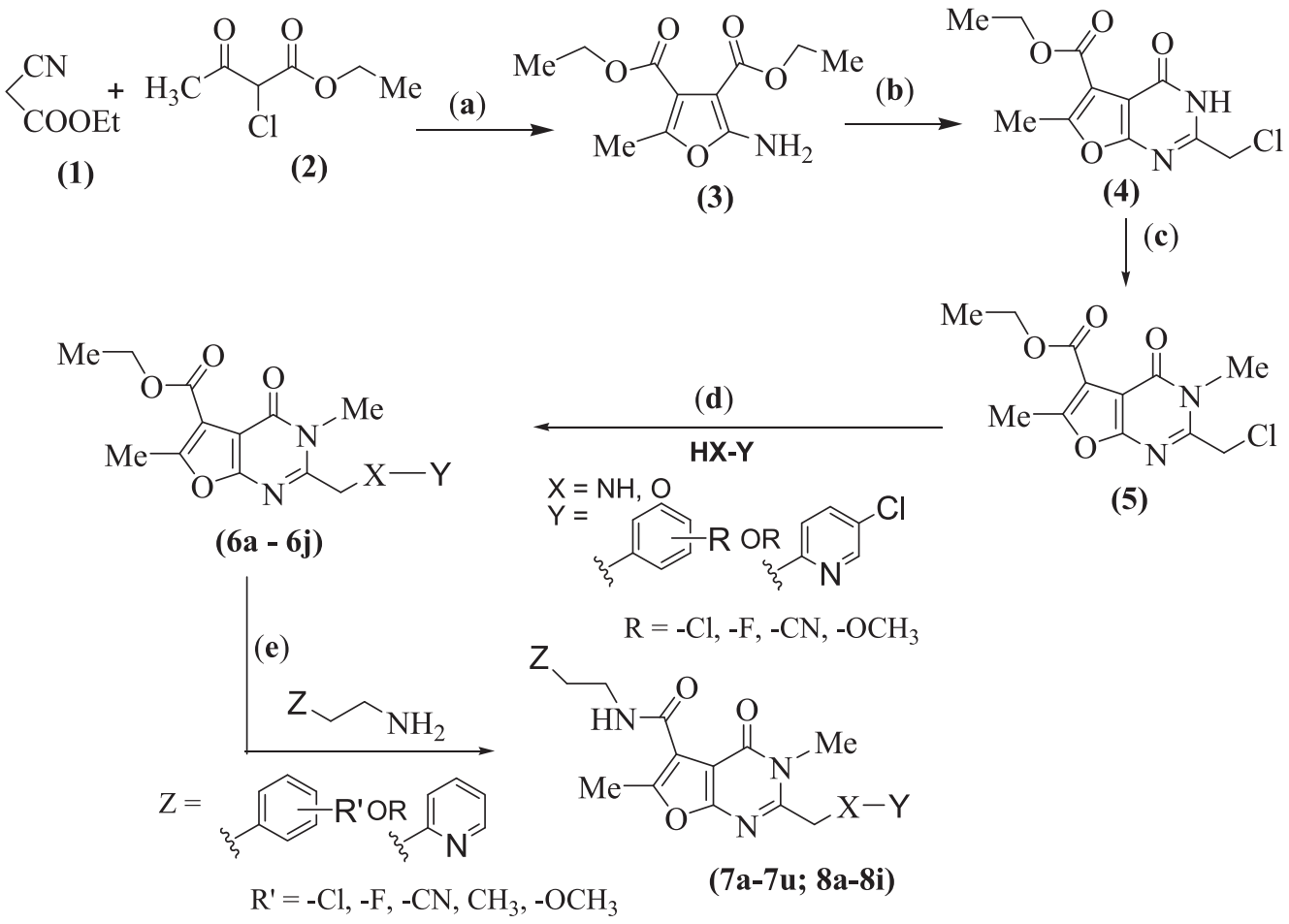
Structural Optimization: Structural modifications were performed for the development of potent novel antithrombotic compounds. The central furano-pyrimidinone scaffold was envisaged from a combination of fragment-based drug discovery (FBDD), bioisosterism and molecular hybridization approaches. This scaffold possesses π-π stacking interactions with Tyr60A-Trp60D pocket of the S2-binding site, that is crucial for thrombin activity. Modifications were carried out on both P1

Table 1
IC_{50} values of the selected target compounds for thrombin inhibition.



(7a -7u; X = NH)
(8a - 8i; X = O)

Comp	Y	Z	*IC_{50} (μM)	Comp	Y	Z	*IC_{50} (μM)
7a	4-ClPh	Ph	31.2 ± 5.6	**7q**	4-CNPh	4-FPh	31.2 ± 5.6
7b	4-ClPh	4-FPh	**0.96 ± 0.2**	**7r**	4-CNPh	4-ClPh	23.6 ± 3.7
7c	4-ClPh	4-OMePh	21.7 ± 4.8	**7 s**	4-CNPh	4-OMePh	52.2 ± 6.3
7d	4-ClPh	4-MePh	ND	**7 t**	4-OMePh	4-FPh	68.6 ± 8.1
7e	4-ClPh		**1.43 ± 0.3**	**7u**		4-MePh	21.4 ± 4.3
7f	4-FPh	Ph	6.53 ± 1.3	**8a**	4-CNPh	Ph	ND
7g	4-FPh	4-ClPh	**1.36 ± 0.4**	**8b**	4-CNPh	4-FPh	47.3 ± 4.4
7h	4-FPh	4-OMePh	25.2 ± 4.7	**8c**	4-CNPh	4-ClPh	38.7 ± 3.6
7i	4-FPh	4-MePh	18.8 ± 2.6	**8d**	4-CNPh	4-MePh	42.6 ± 5.2
7j	3-Cl,4-FPh	Ph	8.69 ± 1.2	**8e**	4-CNPh		25.4 ± 3.8
7 k	3-Cl,4-FPh	4-ClPh	3.01 ± 0.7	**8f**	4-ClPh	Ph	55.4 ± 7.8
7 l	3-Cl,4-FPh	4-FPh	4.55 ± 0.8	**8 g**	4-ClPh	4-FPh	ND
7m	3-Cl,4-FPh	4-OMePh	ND	**8 h**	4-FPh	Ph	ND
7n	3-Cl,4-FPh	4-MePh	ND	**8i**	4-FPh	4-ClPh	ND
7o	4-CNPh	Ph	34.6 ± 6.3	**Dabigatran**		0.058 ± 0.006	
7p	4-CNPh	4-MePh	ND	*The IC_{50} values expressed as mean ± SD (n = 3) ND = not determined			



Scheme 1. Synthesis of compounds (**7a-7u** & **8a-8i**). Reagents and conditions: (a) TEA, iPrOH, 4 h.; (b) Chloroacetonitrile, Dry dioxane in Dry HCl, 0–5 °C; (c) DMS, Sodium bicarbonate, Dry acetone, reflux; (d) Dry dioxane, reflux; (e) Trimethyl aluminium (2 M solution in toluene), EDC, RT.

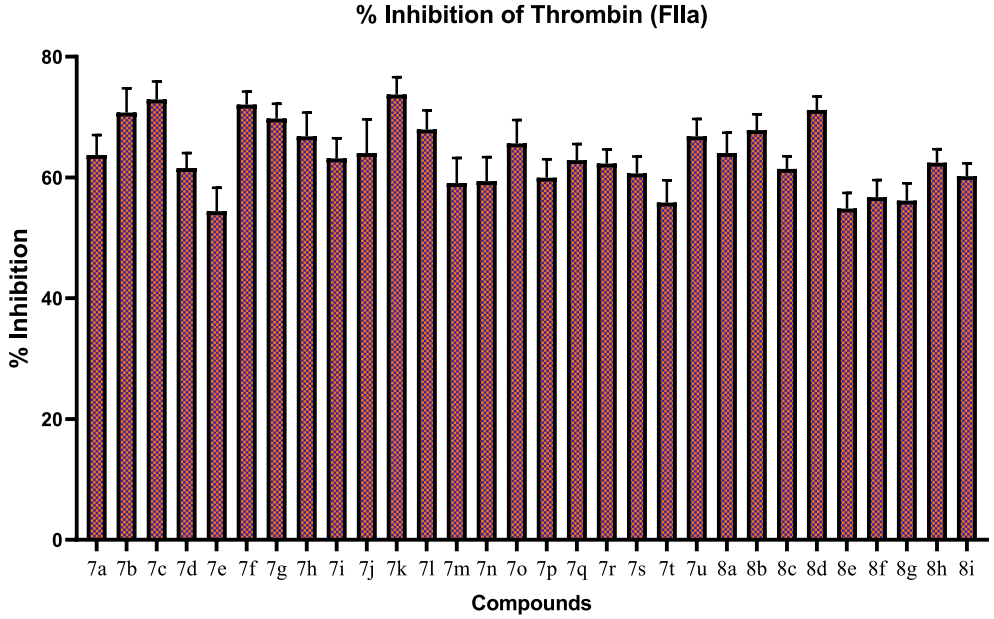


Fig. 3. *In vitro* Thrombin (FIIa) enzyme inhibition activity of the target compounds (**7a-7u; 8a-8i**) at 20 μM concentration.

and P3 moieties to get potential novel antithrombin compounds. Neutral polarizable moieties like small halogen atoms *e.g.* chloro and fluoro and low molecular weight methyl, methoxyl or cyano groups that have hydrophobic and electrostatic interactions with the specific S1 binding site were introduced on P1 side chain. Basic amino group of dabigatran and its isosteric phenolic group were included in the structure to

maintain the biological activity profile. A methyl substituent was incorporated in the central pyrimidone ring that provide stability and enhances potency and selectivity for thrombin enzyme. An ethylene bridge linker was added in P3 side chain to induce S-or U-shaped conformation to the target molecules to accommodate them into the thrombin binding active site (Fig. 4). Aromatic halophenyl or pyridyl fragments binding to lipophilic S3 binding site are accommodated in P3 chain. Highly basic amidine group and non-essential pyridine ring present in dabigatran were done away to improve pharmacokinetics profile and anti-thrombotic potency.

From the inhibitory activities of the synthesized compounds (**7a-7u** and **8a-8i),** it was found out that compounds (**7b,** IC_{50} = 0.96 μM and **7g,** IC_{50} = 1.36 μM), containing 4-chlorophenyl and 4-fluorophenyl as the S1-binding fragments and 4-fluoro- and 4-chlorophenethyl as S3-binding motifs displayed the most potent inhibitory activity. This indicated that electron-withdrawing halogen groups displayed enhanced antithrombotic activity. Compounds having 3-Chloro-4-fluorophenyl P1-fragment (**7k,** IC_{50} = 3.01 μM; **7l,** IC_{50} = 4.55 μM) resulted in moderate biological activity while 4-cyanophenyl compounds (**7o**; IC_{50} = 34.6 μM, **7q**; IC_{50} = 31.2 μM; **7r**; IC_{50} = 23.6 μM, **7s**; IC_{50} = 52.2 μM) exhibited weak potency. Introduction of electron-donating groups like methoxy or methyl on S1-binding fragments conferred compounds (**7c, 7h, 7i** and **7u**) with decreased inhibition. Antithrombotic activity also diminished in the order as 4-chloro >4-fluoro >3-chloro-4-fluoro >4-cyano >4-methoxy, among S1-ocupying substituted aniline derivatives. Replacement of phenyl ring with 5-chloro-2-pyridyl as P1-moiety furnished compound (**7u,** IC_{50} = 21.4 μM) having reduced potency. Compound containing S3-binding 2-pyridyl and S1-targeting 4-haloanilino fragments exhibited potent inhibitory activity (**7e**; IC_{50} = 1.43 μM). Furthermore, modifications of P1-fragments in the form of phenolic linker moieties did not improve the biological activity (**8b**; IC_{50} = 47.3 μM; **8c**; IC_{50} = 38.7 μM, **8f**; IC_{50} = 55.4 μM).

3.2.3. *Ex vivo evaluation of prothrombin time (PT) and activated partial thromboplastin time (aPTT) prolongation*

The synthesized analogs that displayed significant biological activity from *in vitro* inhibition studies were next assessed for the anticoagulant profile by estimating *ex vivo* prothrombin time and activated partial thromboplastin time prolongation.[57] PT represents the effect of the inhibitor on extrinsic pathway, while aPTT indicates inhibitor's influence on intrinsic pathway of coagulation.[58] The activity was performed on rats at oral dose of 30 mg/kg of the test compounds. The results (Table 2) revealed that derivatives (**7b,** PT = 18.7 s and aPTT = 33.4 s and **7l,** PT = 11.7 s and aPTT = 35.3 s) showed more prolonged anticoagulation than the control (PT = 8.7 and aPTT = 25.3 s) but comparatively lesser than the standard drug dabigatran (PT = 26.1 and aPTT = 101 s). Compounds **7g** (containing 4-chlorophenyl P3-fragment), **7c** (containing 4-methoxyphenyl P3-fragment) and **7e** (containing 2-pyridyl P3-fragment) exhibited weak PT/aPTT prolongation activity. Both **7c** and **7e** showed weaker thrombin enzyme binding interactions in docking studies. While **7g** though displayed potent *in vitro* thrombin inhibition, was found as an exception**.**

Table 2
Ex vivo evaluation of PT and aPTT prolongation and bleeding time in rats of some selected compounds.

Compound	PT (sec.) (30 mg/kg dose)	aPTT (*sec.*) (30 mg/kg dose)	Bleeding time** (sec.) (5 mg/kg dose)
7a	10.6	27.5	122 ± 3.7**
7b	**18.7**	**33.4**	**91** ± 2.6**
7c	9.2	19.2	156 ± 4.8**
7e	11.5	18.9	145 ± 4.1**
7f	**11.5**	**29.9**	106 ± 2.7**
7g	9.9	18.5	135 ± 3.5**
7i	10.3	22.3	128 ± 3.2**
7j	9.8	20.6	165 ± 5.3**
7k	**13.2**	27.4	136 ± 4.1**
7l	11.7	**35.3**	124 ± 3.4**
Dabigatran	**26.1**	**101**	**102** ± 2.5**
Control	8.7	25.3	78 ± 2.2**

Test results are expressed in seconds as mean ± S.E.M. Statistical significance between two groups as $^{**}P < 0.05$.

3.2.4. *Ex vivo bleeding time in rats*

Selected compounds were also screened for the bleeding time measurement to assess their potential for hemorrhagic tendency.[59] The results (Table 2) demonstrated that bleeding tendency of the selected target compounds was slightly higher than the standard drug at the antithrombotic dose (30 mg/kg) except for compounds (**7c** and **7j**) wherein the time was significantly higher. Interestingly, compound (**7b)** exhibited lower bleeding time (BT = 91 s.) compared to the standard drug dabigatran (102 s.).

3.3. *Molecular docking studies*

To start docking studies, the thrombin crystal structure was prepared

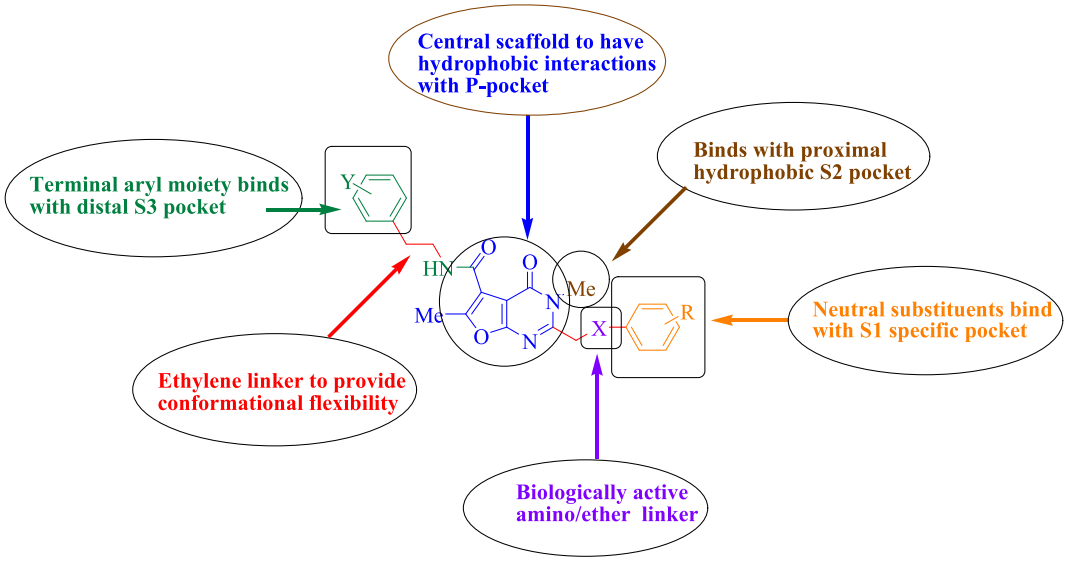


Fig. 4. Binding interactions of the target compounds in the thrombin binding site.

by running a 100 ns molecular dynamics simulation in explicit solvent to optimise the receptor and alleviate strain energies. Further details are available in the MDS section. The receptor complex from the final frame of the 100 ns MDS was used for docking experiments. The study was validated by redocking of the native ligand on the pre and post simulated protein structure, the redocked results showed a nice fit to the native crystal structure pose of dabigatran (RMSD = 1.024 Å) (**SI, Fig. S1**).

The target compounds were examined for molecular interactions with thrombin. The docking scores are displayed in Table 3. The overlaid structures of the most active compound (**7b**) and dabigatran shared close binding interactions with thrombin as reported earlier.[60] The molecular binding of compound (**7b**) with the thrombin enzyme is shown in Fig. 5 **(a-c)**, where the native ligand and **7b** showed a good overlay in the binding site. In its docked form, compound **7b** exhibits an extended, U-shaped conformation shaped by the central *N*-methyl furo [2,3-*d*]pyrimidinone core and the ethylene-bridge linker (**Fig. S3**). This shape allows the molecule to simultaneously interact with all three main thrombin subsites - the neutral polarizable 4-chlorophenyl P1 fragment extends into the S1 pocket toward Asp199, while the methylated pyrimidinone core fits into the aromatic S2 pocket, stacking π-interactions with Tyr47. The *N*-methyl group stabilizes the lactam and enhances thrombin selectivity. The terminal 4-fluorophenethyl group folds into the lipophilic S3 groove, lined by Leu96 and Ile179. Its electronegative *para*-fluoro and chlorophenyl rings participate in halogen-related hydrophobic and electrostatic interactions, which are crucial as the SAR data shows. The binding maintains a network of polar contacts, including a strong hydrogen bond at 2.79 Å with the lactam/amide core, and additional interactions at about 3.03–3.14 Å with Asn95, approximately 3.39 Å with Leu96 backbone, around 3.42 Å with Trp92 backbone carbonyl, a weaker contact near 3.57 Å with Glu229, and a hydrogen bond to Tyr47. Some important van der Waals interactions were also reported with the essential residues of the enzyme, *e.g.* Glu97A, Glu192, Ala190, Ser214, Gly216. Other compounds analysed for molecular docking showed similar binding conformations and docking scores. *Ex vivo* tests identified compounds (**7f** and **7 l**) as lead candidates based on their biological profiles. Their docking scores (−8.07 and − 8.34, respectively) are comparable to that of compound (**7b,** −8.49). All these scores are somewhat poorer than that of the native ligand, dabigatran, which is −9.33. This outcome was anticipated, given that the co-crystal structure served as the protein receptor in this study. The consistent results from docking experiments and biological evaluations encouraged us to conduct molecular dynamics simulations for the selected ligands. This step aims to confirm their binding efficiency to the target receptor, assess the stability of the complexes, and measure the molecules' affinity for the binding site.

Building on the existing SAR, which indicates that potency correlates with the electron-withdrawing and polarizable nature of the P1 aryl (4-Cl > 4-F > 3-Cl,4-F > 4-CN > 4-OMe), the anilino series consistently outperforms its phenolic-ether counterparts (**8b/8c/8f**). Moreover, 4-halophenethyl or 2-pyridyl P3 groups are optimal. MM-GBSA disposition shows that the most significant change is in compound **7b**, where its van der Waals energy (−50.8 kcal/mol) nearly matches dabigatran's (−51.2 kcal/mol). However, its electrostatic contribution is much weaker (−21 *vs.* -172), suggesting that shape complementarity is maximized and the ~16-fold potency difference mainly stems from a lack of electrostatic attraction at S1. This shortfall occurs because dabigatran's highly basic amidine was replaced with a less basic group to improve pharmacokinetics. The aim is to reintroduce a neutral yet effective S1 anchor, such as an amidine bioisostere like aminomethyl, methyl-sulfonamide, aminopyridine, or a neutral chlorothiophene motif, or a heavier σ-hole halogen capable of forming a directional H-bond, salt bridge, or halogen bond with Asp199, without increasing basicity and impairing absorption. Our current approach preserves the most potent compound **7b**'s positive ADMET profile, Lipinski compliance, over 90% intestinal absorption, and absence of blood-brain barrier penetration.

Table 3
Docking scores of the synthesized compounds.

Compound	Docking score	Compound	Docking score
7b	−8.49	**7k**	−7.59
7e	−7.34	**7l**	−8.34
7f	−8.07	**8e**	−7.15
7g	**−9.35**	**Dabigatran**	**−9.33**
7j	−7.76		

3.4. Molecular dynamics simulation (MDS) study

The MDS of the target protein in Apo- (Unbound state of thrombin protein) and bound forms were conducted using the method outlined in the experimental section. All trajectories were stripped of water molecules and analysed to evaluate intermolecular interactions and the RMSD of the protein receptor and ligands relative to the initial bound state. Additionally, RMSF was calculated to assess residue fluctuations and the impact of ligand binding on the binding site. The native ligand, dabigatran, initially formed hydrogen bonds with Asp199, Cys231, and Gly230 during the early stage of the MDS. Among these, Asp199 and Gly230 were maintained throughout the simulation (**Fig. 6a** and **b, Fig. S2**). Additional hydrogen bonds with Arg178 and Glu202 were observed initially, but only Arg178 persisted until the end of the MDS. The protein RMSD for the dabigatran-bound protein showed an average value of 0.93 Å (Table 4). The protein structure stabilised around 25 ns and remained consistent throughout rest of the simulation (Fig. 8). The ligand RMSD for dabigatran remained low during the MD simulation, with fluctuations under 1 Å and an average of 0.90 Å over the entire period (Fig. 9). RMSF analysis of the protein residues revealed that the dabigatran-bound form of thrombin consistently exhibited lower RMSF values compared to the Apo-form (Black) (Fig. 7). The dabigatran-thrombin enzyme complex showed the highest binding energy (ΔG) of −59.28 ± 5.51 kcal/mol. Compound (**7b**) initially interacted with Glu94, Asn95, and Trp92 at the start of MD simulations and continued to form interactions, including a hydrogen Bond with Tyr47 toward the end (**Fig. 6c** and **d**). The ligand remained stable throughout the simulation period, as indicated by a ligand RMSD of 0.69 Å and a protein RMSD of 0.95 Å.

It stabilised the protein-binding site more effectively than the other investigated ligands. The ligand RMSD was comparable to that of dabigatran (Fig. 9**, red**). The **7b**-thrombin complex exhibited a binding energy (ΔG) of −37.98 ± 3.55 kcal/mol. During the MDS, compound (**7f**) initially formed a hydrogen bond with Cys231 and a halogen interaction with Trp227. After completing the simulation, **7f** established hydrogen bonds with Ala200 and Arg178 (**Fig. 6e** and **f**). The Ligand RMSD fluctuated throughout the MDS, with noticeable increases around 40 ns and 60 ns, averaging 0.85 Å - the highest among all studied compounds. These fluctuations suggest conformational changes during the simulation (Fig. 9**,** green). The protein RMSD was slightly higher, averaging 1.10 Å compared to the other ligand-protein complexes. The **7f**-thrombin complex exhibited a binding energy (ΔG) of −38.60 ± 3.59 kcal/mol. Compound (**7l**) initially formed Hydrogen bonds with Leu96, Asn95, and Trp92 before MDS, then established new interactions with Asn95, Tyr47, Gly228, and Cys28 (**Fig. 6g** and **h**). The protein RMSD averaged 1.03 Å and stabilised around 50 ns (Fig. 8). The ligand RMSD was lower, averaging around 0.40 Å, with the lowest values at 60 ns (Fig. 9**,** Table 4). The protein RMSF was similar to that of other complexes, and the ligand exhibited lower fluctuations than in the Apo protein structure (Fig. 7). The **7l**-thrombin complex showed a binding energy (ΔG) of −33.05 ± 3.51 kcal/mol, the lowest among the tested ligands (Table 4). The molecular docking and molecular dynamics simulations suggest that these ligands stabilise the thrombin protein and do not diffuse out of the binding site, making them effective thrombin

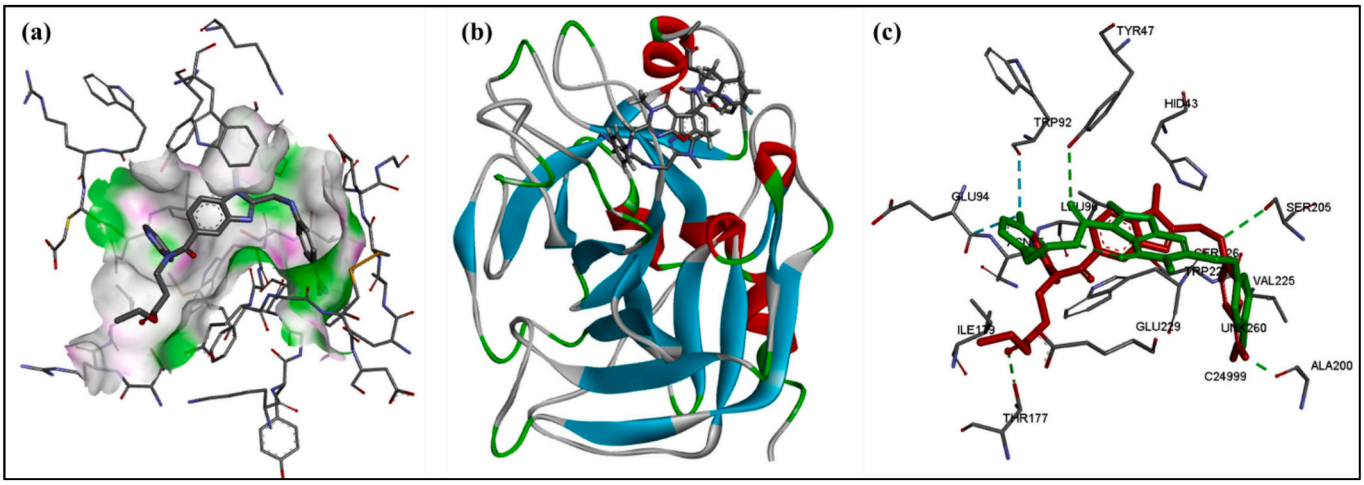


Fig. 5. Molecular docking studies (**a**) binding site for the native ligand dabigatran in the Thrombin Protein (PDB: 1KTS); (**b**) Protein-ligand complex of thrombin protein with dabigatran and compound **7b**, and (**c**) 3D ligand overlay and the interaction pose for compound **7b** (green) with thrombin enzyme. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

binders.

3.5. Prediction of drug-likeliness and pharmacokinetics properties

The drug-likeliness and pharmacokinetics parameters of compounds are contemplated as the indispensable aspects in drug discovery process. The drug-likeliness and ADMET properties of the target compounds were computed from the online tools, SwissADME and pkCSM.[61,62] The results are shown in Table 5. Herein, the target compounds followed drug-likeness properties as per Lipinski's rule of five without violation. With more than 90% intestinal absorption for all these compounds indicated their oral efficacy. Not a single compound exhibited BBB permeability thus implying absence of CNS toxicity. The excretion and toxicity prediction data was also reported favorable for drug-like properties with total clearance within the range, and without hERG inhibition and AMES mutagenicity (except for compound **8e**). Overall, these findings indicated that the final reported compounds have excellent lead potential for their development as oral antithrombotic agents.

4. Conclusion

With the aim of discovering and developing novel direct thrombin inhibitors, a new series of furo[2,3-*d*]pyrimidinone derivatives were designed using the principles of bio-isosterism and molecular hybridization. The designed compounds were synthesized and evaluated for their direct thrombin inhibition, selectivity over FXa, and anticoagulant and bleeding risk properties. Compounds **(7b** and **7g)** containing electron-withdrawing groups displayed potent activity against thrombin having IC_{50} values of 0.96 μM and 1.36 μM respectively. Additionally, compound (**7b)** exhibited favorable *ex vivo* anticoagulant activity in rats along with better safety profile toward bleeding risk compared to the standard. Computed physicochemical properties of the target compounds revealed their drug-like characteristics. Molecular docking studies gave an early insight into the binding affinity of the designed molecules toward the thrombin receptor. This was further confirmed by molecular dynamics simulation studies, in which the protein-ligand complex remained stable throughout the simulation period, and none of the ligands diffused out of the binding site, suggesting favorable interactions. The binding free energy of dabigatran exhibited the most favorable ΔG (−59.28), indicating the strongest predicted affinity for the target. The gap between standard drugs and experimental molecules is large. Ligands 7b and 7f showed comparable binding, with a slight difference but within the expected uncertainty of the computational methods. However, the synthesized ligands showed a similar binding profile, which is a lead attribute for identifying them as potential thrombin inhibitors. Overall, our findings revealed the potential of furano-pyrimidinone structure as a novel scaffold for the development of anti-thrombin compounds, and afford a promising lead that could be further investigated.

5. Future directions

In optimization of our designed compounds we aimed at retaining potent antithrombotic activity and maintaining the improved bleeding profile. The future lower-risk designing strategies may include reinforcing the halo functions like 3,4-difluoro/3,4-dichloro/3,5-dichloro/3-trifluoromethyl, 5-chloro/4,5-dichloro group *etc.* on P1 as 3-Cl and 4-F remained active, and keeping the aniline N—H over a phenolic ether to maintain hydrogen bond donor properties. Lipophilic S3 pocket may be filled with aromatic bulk groups, or directing a polar handle toward Glu94/Glu229 to capture charged interactions. Rotatable bonds may be reduced by fixing the ethylene-bridge linker *via* ring fusion or macrocyclization to lower conformational entropy, eliminating CYP3A4 soft spots, and addressing electrostatic, entropic, or metabolic weaknesses. In future, a detailed structural optimization strategy may be adopted in order to achieve enhanced relevant *in vivo* biological activity.

6. Experimental

6.1. Chemical work

All the chemicals and reagents were commercially purchased and used as such without purification unless required. The synthetic procedures were monitored by thin-layer chromatography (TLC) using precoated silica gel G plates. The synthesized compounds were purified either by recrystallization or column chromatography. Melting points were determined on programmable melting point instrument and are unedited. FT-IR was used to record the IR spectra (in cm^{-1}) using KBr pellets. ^{1}H NMR/^{13}C NMR spectra were recorded on 400 MHz/100 MHz spectrometer with TMS as internal standard. Mass spectra were acquired on instrument with EI ion source or micrOTOF-Q II spectrometer with ESI source. Elemental analyses were carried out on GC-based elemental analyzer. The thrombin enzyme along with the substrate were obtained commercially.

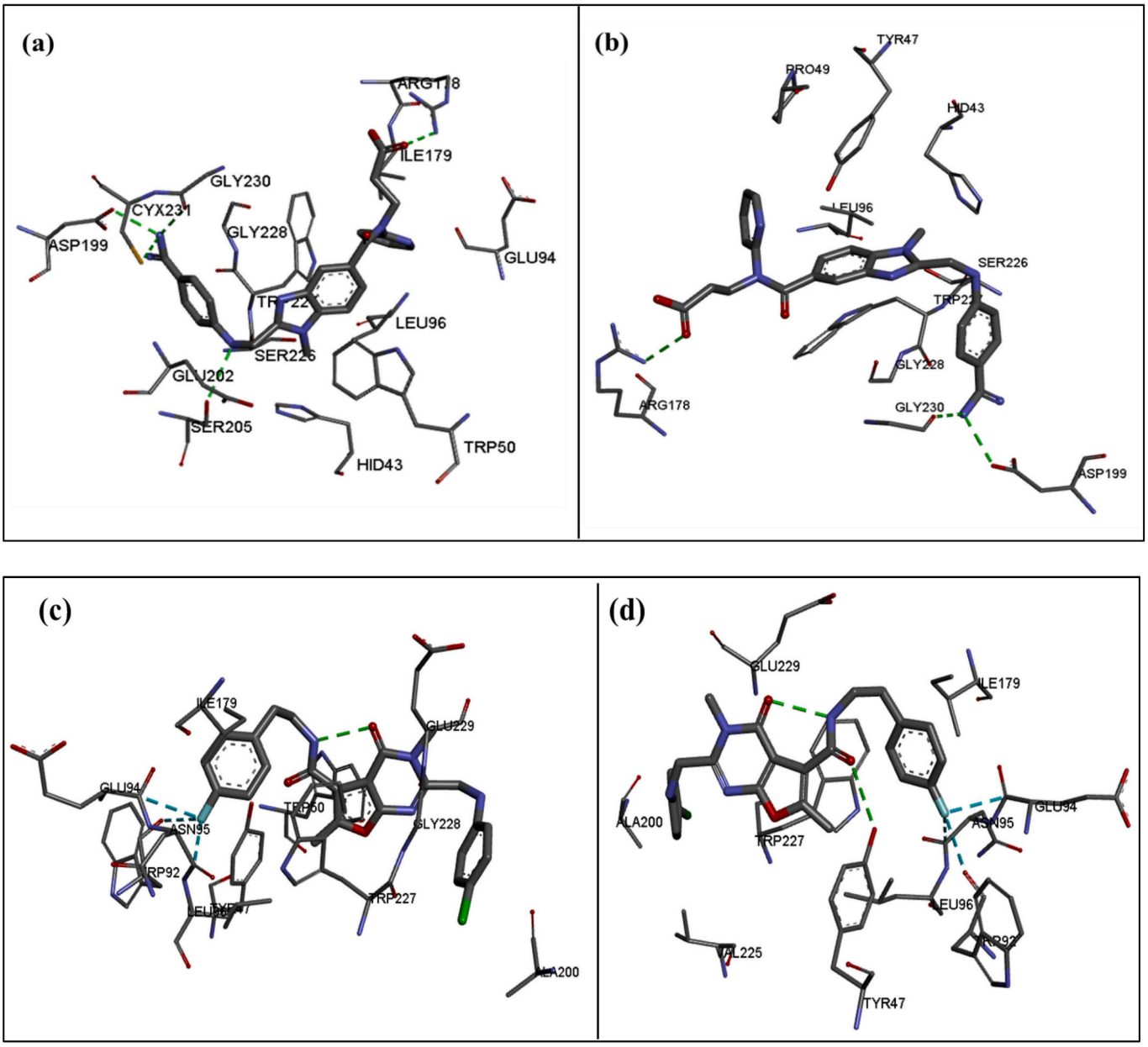


Fig. 6. Molecular dynamics simulations (MDS) of thrombin enzyme with the reference ligand dabigatran and synthetic compounds, the MDS initial and final stages following 100 ns; (**a**, **b**) Dabigatran; (**c**, **d**) **7b**; (**e**, **f**) **7f** and (**g**, **h**) **7 l**.

6.2. Diethyl 2-amino-5-methylfuran-3,4-dicarboxylate (3)

A mixture of ethyl cyanoacetate (**1**) (1.0 g, 0.94 ml, 8.84 mmol) and triethylamine (1.78 g, 17.68 mmol) in isopropyl alcohol (8 ml) was stirred at room temperature for 10–20 min. Ethyl 2-chloroacetoacetate (**2**) (1.45 g, 8.84 mmol) was added into the above solution slowly with continuous stirring at cold condition (0–5 °C). The reaction mixture was then stirred at room temperature for 3–4 h and monitored by TLC. Crushed ice was added into the reaction mixture to get a solid that was filtered and washed with cold water. The crude product (**3**) was dried and used as such for next step. Yellow solid, **M.p.** 82–83 °C. **Yield:** 80%. **IR**: 3429, 3325, 1708, 1673, 1554 and 1219.

6.3. Ethyl 2-(chloromethyl)-3,4-dihydro-6-methyl-4-oxofuro[2,3-d] pyrimidine-5-carboxylate (4)

Diethyl 2-amino-5-methylfuran-3,4-dicarboxylate (**3**) (1.0 g, 4.14 mmol) and chloroacetonitrile (0.93 g, 12.43 mmol) were mixed and stirred at room temperature for half an hour. In this mixture, dry HCl gas in dry dioxane solution (6 ml) was slowly added at cold condition (0–5 °C) with constant stirring that led to a clear orange solution which was further stirred for 5–6 h and monitored by TLC. At the end of the reaction ice crystals were added into the mixture to get the precipitate. Liquor ammonia was added to neutralize excess HCl. The solid so obtained was filtered under suction to get the product (**4**). Off White solid, **M.p.** 201–203 °C. **Yield:** 70%. **IR**: 1246, 1310, 1676, 1719 and 3430. **^{1}H NMR** (400 MHz, DMSO-d_6): δ 1.34 (t, 3H, $J = 7.0$ Hz, -CH_2CH_3), 2.63 (s, 3H, Ar-CH_3), 4.28–4.32 (q, 2H, $J = 7.0$ Hz, -CH_2CH_3), 4.53 (s, 2H, -CH_2Cl), 12.24 (bs, 1H, -N*H*), **MS** (*m/z*): 271 $[M]^+$, 273 $[M + 2]^+$.

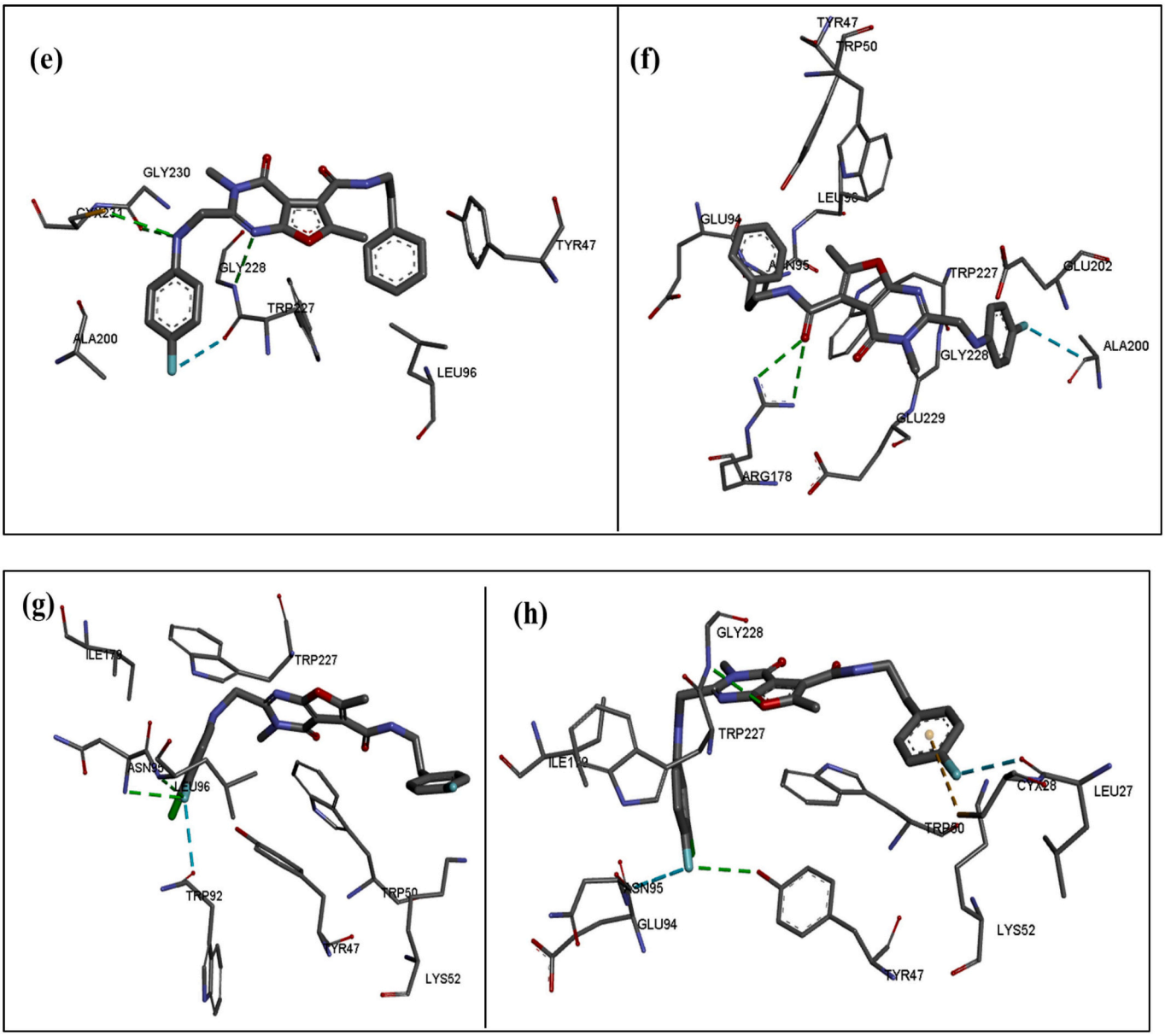


Fig. 6. (*continued*).

6.4. *Ethyl 2-(chloromethyl)-3,4-dihydro-3,6-dimethyl-4-oxofuro[2,3-d] pyrimidine-5-carboxylate (5)*

Ethyl 2-(chloromethyl)-1,4-dihydro-6-methyl-4-oxofuro[2,3-*d*]pyrimidine-5-carboxylate, (**4**) (1.0 g, 3.69 mmol), sodium bicarbonate (0.62 g, 7.38 mmol) and dry acetone (30 ml) were mixed and stirred at room temperature under anhydrous conditions in a RBF for about half an hour. Dimethyl sulphate (0.69 g, 5.54 mmol) was introduced dropwise into the mixture at cold conditions and then refluxed at 60–65 °C for 6–7 h under nitrogen and monitored. The crude solid was obtained by addition of crushed ice to the reaction mixture. The precipitate was filtered under suction and washed with cold water to obtain compound (**5**). Light Brown solid. **M.p.** 120–122 °C. **Yield:** 62%. **IR**: 1223, 1350, 1547, 1581, 1689 and 1727. **^{1}H NMR** (400 MHz, DMSO-d^6): δ 1.41 (t, 3H, J = 7.2 Hz, -CH_2CH_3), 2.66 (s, 3H, Ar-CH_3), 3.72 (s, 3H, -NCH_3), 4.40 (q, 2H, J = 7.1 Hz, -CH_2CH_3), 4.59 (s, 2H, -CH_2Cl), **MS** (*m/z*): 284 $[M]^+$, 286 $[M + 2]^+$.

6.5. *Ethyl 2-((4-chlorophenylamino)methyl)-3,4-dihydro-3,6-dimethyl-4-oxofuro[2,3-d] pyrimidine-5-carboxylate (6a)*

Ethyl 2-(chloromethyl)-1,4-dihydro-3,6-dimethyl-4-oxofuro[2,3-*d*] pyrimidine-5-carboxylate (**5**) (1.0 g, 3.51 mmol) and 4-chloroaniline (0.89 g, 7.02 mmol) were mixed in dry dioxane (20 ml) was refluxed overnight and was monitored by TLC. Finally, ice-cold water was introduced into the mixture to get the precipitate that was filtered under suction to get compound (**6a**). Light yellow solid, **M.p.** 185–187 °C. **Yield:** 64% **IR**: 1090, 1586, 1683, 1732 and 3377. **^{1}H NMR** (500 MHz, $CDCl_3$): δ 1.43 (t, J = 7.5 Hz, 3H, CH_2CH_3), 2.68 (s, 3H, ArCH_3), 3.63 (s, 3H, -NCH_3), 4.33 (s, 2H, CH_2NH), 4.39–4.41 (q, J = 7.0 Hz, 2H, CH_2CH_3), 5.11 (s, 1H, -N*H*), 6.66–6.68 (d, J = 8.5 Hz,2H, Ar—*H*), 7.17–7.20 (d, J = 9.0 Hz, 2H, Ar—*H*), **MS** (*m/z*): 376 $[M + 1]^+$, 377 $[M + 2]^+$.

Compounds **6b-6i** were synthesized according to the procedure described above for compound **6a** and reported with spectral data in supporting information file.

Table 4
Molecular docking, MD simulation and binding free energy components for the protein- ligand complexes calculated by MM-GBSA analysis; all energies are in kcal/mol with standard deviation in parentheses.

Enzyme/ ligand	Molecular dynamics simulations			MM-GBSA* (kcal/mol)						
	Dock Score	RMSD (Å)(avg)		ΔE_{VDW}	ΔE_{ELE}	ΔG_{GB}	ΔG_{Surf}	ΔG_{gas}	ΔG_{Sol}	ΔG_{bind}
		Protein	Ligand*							
Thrombin (PDB:1KTS)	–	0.96	–	–	–	–	–	–	–	–
Dabigatran	−9.33	0.93	0.90	−51.19 ± 5.00	−171.80 ± 28.65	168.35 ± 25.89	−6.43 ± 0.41	−221.20 ± 29.34	161.92 ± 25.62	−59.28 ± 5.51
7b	−8.49	0.95	0.69	−50.82 ± 3.25	−20.96 ± 5.16	40.38 ± 4.52	−5.48 ± 0.33	−72.89 ± 6.55	34.90 ± 4.37	−37.98 ± 3.55
7f	−8.07	1.10	0.85	−51.70 ± 3.27	−20.20 ± 7.11	39.79 ± 6.63	−5.58 ± 0.30	−72.82 ± 8.39	34.21 ± 6.48	−38.60 ± 3.59
7l	−8.34	1.03	0.40	−49.22 ± 3.63	−11.50 ± 4.44	34.14 ± 4.46	−5.67 ± 0.33	−61.51 ± 5.66	28.46 ± 4.40	−33.05 ± 3.51

* **RMSD** = With reference to input/reference structure; ΔE_{VDW} = van der Waals contribution from MM; ΔE_{ELE} = electrostatic energy as calculated by the MM force field; ΔG_{GB} = the electrostatic contribution to the solvation free energy calculated by GB; ΔG_{Surf} = solvent-accessible surface area; ΔG_{Sol} = solvation free energy; ΔG_{gas} = gas phase interaction energy; ΔG_{bind} = Binding free energy.

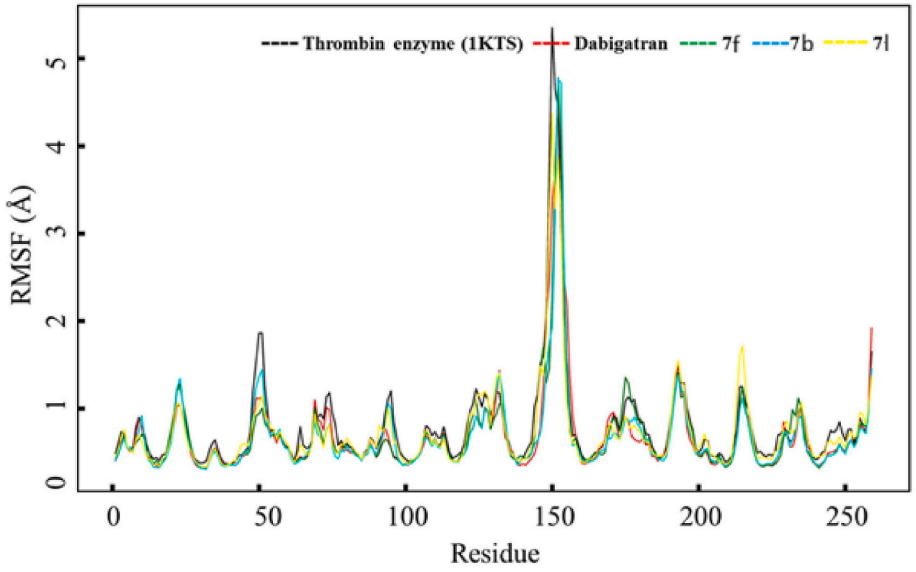


Fig. 7. Protein RMSF of unbound thrombin enzyme and the ligand-bound enzyme through the 100 ns MDS.

6.6. 2-((4-Chlorophenylamino)methyl)-3,4-dihydro-3,6-dimethyl-4-oxo-N-phenethylfuro[2,3-d] pyrimidine-5-carboxamide (7a)

β-Phenethylamine (0.38 g, 3.19 mmol) and ethylene dichloride (3 ml) were mixed under the anhydrous conditions in a dry RBF. Trimethylaluminium, 2.0 M solution in toluene (0.34 g; 2.46 ml, 4.78 mmol) was added dropwise to the above mixture at 0 °C and was stirred at room temperature for half an hour. Then, compound **6a** (0.2 g, 0.53 mmol) in 3 ml ethylene dichloride was added to this mixture and stirred overnight at room temperature. The reaction mixture was quenched by adding 2 M hydrochloric acid and stirred. The mixture was then extracted with methylene chloride and the organic layer was dried over anhydrous sodium sulphate and evaporated. The solid obtained was finally purified using column chromatography to get compound (**7a**). Light brown solid, **M.p.** 178–180 °C. **Yield:** 52%. **IR**: 1251, 1595, 1639, 1665, 2922 and 3251. **^{1}H NMR** (500 MHz, $CDCl_3$): δ 2.85 (s, 3H, Ar-C*H*$_3$), 2.98 (t, 2H, J = 8.0 Hz, -C*H*$_2$-CH$_2$), 3.67–3.70 (m, 5H, -CH$_2$-C*H*$_2$–NH & N-C*H*$_3$), 4.40 (s, 2H, -C*H*$_2$–NH), 6.69–6.71 (d, 2H, J = 8.0 Hz, Ar—*H*), 7.21–7.24 (m, 2H, Ar—*H*), 7.28–7.33 (m, 5H, Ar—*H*), 10.03 (t, 1H, J = 5.0 Hz, -CON*H*). **MS** (m/z): 451.3 $[M + 1]^+$. Anal. calcd. For $C_{24}H_{23}ClN_4O_3$ (%): C, 63.93; H, 5.14; N, 12.43. found: C, 63.86; H, 5.08; N, 12.37.

Compounds **7b-7u** and **8a-8i** were synthesized according to the procedure described above for compound **7a** and reported with spectral data in supporting information file.

6.7. Biological evaluation

Chromogenic enzyme assay kit containing thrombin, substrates and buffers were commercially procured. The thrombin inhibition activity was recorded on microplate reader. Male Wistar rats (approx. 250–300 g) were used for biological activity and the protocol for the study was approved by Institutional Animal Ethics Committee (IAEC) of Faculty of Pharmacy, The Maharaja Sayajirao University of Baroda, Vadodara, Gujarat. All experiments were carried out in accordance with Committee for the Purpose of Supervision of Experiments on Animals (CPCSEA) guidelines (Approval No. MSU/IAEC/2019–20/1929).

6.7.1. In vitro thrombin and FXa inhibition assay

The *in vitro* enzyme inhibition effects of the final compounds against thrombin enzyme were determined, from a commercially obtained chromogenic assay kit containing thrombin, the substrate and buffer (*p*H 7.4).[55,56] The target compounds and the standard drug dabigatran

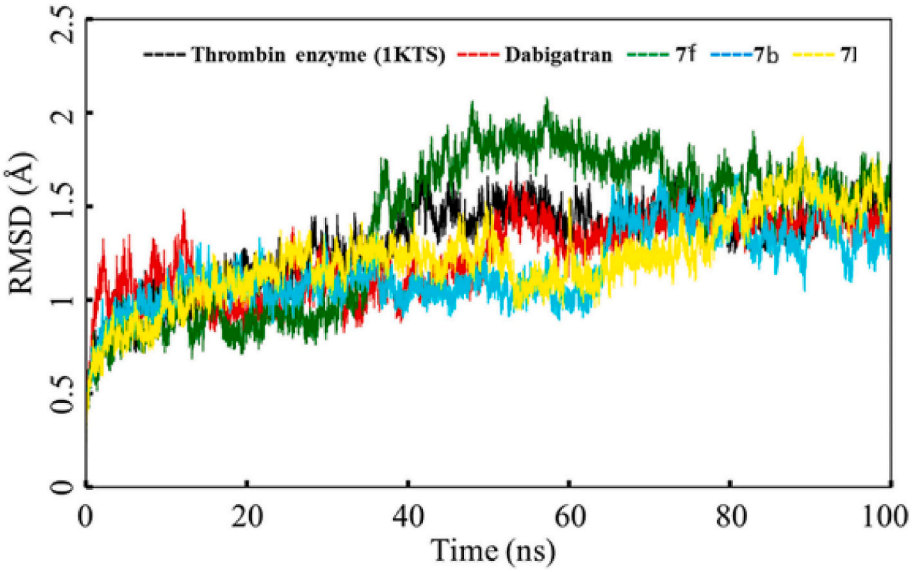


Fig. 8. Protein RMSD of unbound thrombin enzyme and the ligand-bound enzyme through the 100 ns MDS.

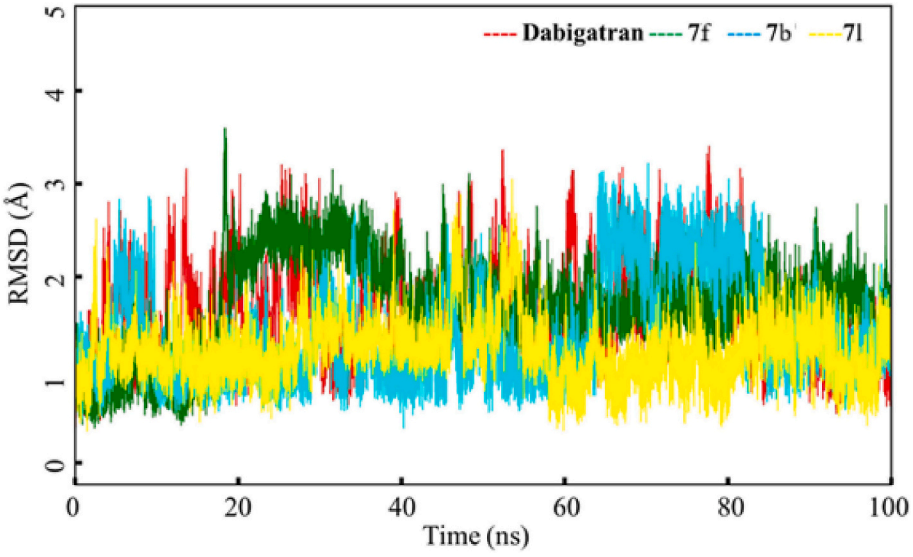


Fig. 9. Ligand RMSD from the ligand-bound enzyme through the 100 ns MDS.

Table 5
Drug-likeliness and pharmacokinetic properties of the target compounds and standard Dabigatran.

Compound	Mol. Wt.	No. of HBD	No. of HBA	Log P	TPSA (A^2)	No. of rotat-able bonds	Caco2 permeability	% Intestinal Absorption (human)	BBB permeability	Metabolism	AMES toxicity	hERG inhibition
7b	468.91	2	5	3.37	89.16	8	1.07	91.49	No	CYP3A4	No	No
7e	451.91	2	5	3.31	102.05	8	1.20	93.52	No	CYP3A4	No	No
7f	434.46	2	5	3.34	89.16	8	1.045	92.91	No	CYP3A4	No	No
7g	468.91	2	5	3.81	89.16	8	0.678	91.25	No	CYP3A4	No	No
7j	468.91	2	6	4.21	89.16	7	0.70	91.24	No	CYP3A4	No	No
7k	503.35	2	5	3.72	89.16	8	1.17	92.14	No	CYP3A4	No	No
7 l	486.90	2	6	3.65	89.16	8	1.38	94.97	No	CYP3A4	No	No
8e	443.45	1	7	3.25	123.04	8	0.53	95.24	No	CYP3A4	Yes	No
Dabigatran	471.51	4	6	1.79	150.22	10	−0.83*	58.73	No	CYP3A4	No	No

HBD = Hydrogen Bond Donor; HBA = Hydrogen Bond Acceptor; TPSA = Total Polar Surface Area; BBB = Blood Brain Barrier, *Caco2 permeability of Dabigatran.[63]

were dissolved in DMSO and diluted with buffer to get concentrations of 0.1 μM - 100 μM solutions. An enzyme assay was performed in 96-well microtiter plates as per the manufacturer's guidelines and recorded on Versa max microplate reader. The test compound dilutions with

thrombin enzyme and buffer were mixed, centrifuged and incubated at 37 °C. Then, enzyme reaction was initiated by addition of thrombin specific substrate and monitored for 20 min. at 405 nM. Direct thrombin inhibitor dabigatran was used as positive control. After initial screening of percentage inhibition of the enzyme by the test compounds at a fixed concentration of 20 μM, the compounds which displayed >60% inhibition were selected for determining the IC_{50} values calculated from GraphPad Prism 5. *In vitro* evaluation of the target compounds for FXa inhibition activity was performed by following the same procedure described for thrombin as above.

6.7.2. *Ex vivo prothrombin time (PT) and activated partial thromboplastin time (aPTT) prolongation in rats*

Prothrombin time (PT) and activated partial thromboplastin time (aPTT) were measured from platelet-poor plasma of male Wistar rats.[57] The synthesized compounds and the standard dabigatran were formulated in 0.5% SCMC (sodium carboxymethylcellulose) and given orally to the animals at 30 mg/kg. After two hours of dosing, blood was collected from retro-orbital plexus in a tube containing sodium citrate. Platelet-poor plasma was collected after centrifugation to determine PT and aPTT.

6.7.3. *In vivo tail-bleeding test*

Bleeding times were determined in healthy male Wistar rats anesthetized *via* intraperitoneal injection of pentobarbital sodium (60 mg/kg) by the method reported previously.[64] The target compounds and standard dabigatran etexilate suspended in sodium carboxymethylcellulose (0.5%) were administered *via* oral route to overnight fasted animals 90 min. Before their tails were transected 4 mm from the tip. The time elapsed until continuous bleeding stopped for more than 10 s. was recorded.

6.8. *Molecular docking and simulation studies*

The 3D structure of the thrombin-dabigatran complex (PDB ID: 1KTS) was retrieved from the Protein Data Bank (RCSB-PDB).[60] This structure was chosen because it includes a known active site with dabigatran bound. To prepare the crystal structure for docking, breaks were repaired, and energy minimization was conducted to relieve strain from crystallization. Protonation was carried out using MOE 2019.01.[65] A dataset comprising dabigatran and the three most potent compounds (**7b**, **7f**, and **7 l**) was used to generate possible conformations *via* the Omega module of OpenEye software.[66] A grid was generated using the 'make receptor' module, with the bound standard drug as the reference in the active site. Molecular docking was then performed with FRED.[67] The top poses with the highest docking scores were retained for further analysis.

6.9. *Molecular dynamics (MD) and molecular mechanics/generalized born surface area (MM/GBSA) studies*

Molecular dynamics simulations were performed using AMBER18 and AmberTools18 with the explicit solvent model.[68,69] Binding energies were calculated using the Molecular Mechanics/Poisson-Boltzmann Surface Area and Molecular Mechanics/Generalized Born Surface Area methods in AMBER18 across the entire trajectory.[70,71] Details of the system setup and unbiased MD simulations are provided in the supplementary information.

CRediT authorship contribution statement

Atul N. Khadse: Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Hardik H. Savsani:** Methodology, Investigation, Data curation, Conceptualization. **Rupesh V. Chikhale:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation. **Shabnam Khan:** Methodology, Data curation. **Shrikant V. Joshi:** Methodology, Data curation. **Prashant R. Murumkar:** Investigation, Formal analysis. **Kirti V. Patel:** Methodology, Investigation, Formal analysis. **Sadhana J. Rajput:** Methodology, Investigation, Formal analysis. **Mange Ram Yadav:** Writing – review & editing, Methodology, Formal analysis, Data curation, Conceptualization.

Ethics declaration

Animal subject

This study was conducted in accordance with the following guidelines for animal welfare and/or reporting: Committee for Control and Supervision of Experiments on Animals (CCSEA), India.

This study was approved by the Institutional Animal Ethics Committee (IAEC), Faculty of Pharmacy, The Maharaja Sayajirao University of Baroda, Vadodara, Gujarat, India. (Approval No. MSU/IAEC/2019–20/1929).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bmc.2026.118764.

Data availability

Data will be made available on request.

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